COCO

CAR SCENTS | ROOM DIFFUSERS | ACCESSORIES

E-Commerce Platform

Technical Analysis & Development Plan

Customer storefront | Owner admin panel | CRM | Inventory | Analytics

Little Scents. Big Vibes.

Prepared by the lead developer | 30 October 2026 | Version 1.0

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OVERVIEW

1. Executive summary

COCO needs two connected products: a **premium, mobile-first storefront** for car scents, room diffusers, car accessories and gift sets, and an **owner admin panel** that runs the whole business: orders, customer database, stock, promotions, reviews and growth analytics including profit. The brief is detailed and explicitly asks for features that standard e-commerce platforms only offer through paid plugins or custom work: swipeable photo/video galleries on product cards, CRM segmentation, profit reporting with product costs, JazzCash/Easypaisa with real payment verification, moderated photo/video/voice reviews, and audit logs.

Recommendation. Build a custom, owner-controlled platform on **Next.js 16 + Payload CMS 3 + PostgreSQL**, deployed as one application. Payload generates a polished, no-code admin panel from our data model, so the owner manages products, orders and content without a developer, while every custom requirement (analytics, CRM, payment verification, media galleries) is built natively instead of bolted on with plugins.

Delivery	Build effort	Recurring cost	Ownership
~12 weeks to launch +2 weeks hypercare	1 senior full-stack dev + part-time UI designer & QA	approx. USD 10-70 / month (no plugin licences)	100% client-owned code, data, domain & accounts

Chosen stack in one line
Next.js storefront + Payload admin in one TypeScript codebase, PostgreSQL database, Cloudflare R2 for media, deployed on a low-cost VPS (or Vercel + Neon for a fully managed option). The project skeleton, data model and core order/CRM/stock logic have already been set up in this repository (see section 18).

Why not WooCommerce or Shopify?

- **WooCommerce** covers the basics but needs 12-18 plugins for this brief (analytics, CRM, swipe galleries, review media, COD rules, invoices, 2FA, backups, exports). Plugin sprawl is what the client explicitly wants to avoid, and it hurts speed on mobile data.
- **Shopify** has no first-party JazzCash/Easypaisa support for Pakistan, charges USD subscriptions plus app fees, and limits custom analytics, profit reporting and data ownership.
- A custom build costs more up front but has the lowest running cost, no licence lock-in, and meets every requirement natively. Full comparison in section 4.

WHAT THE CLIENT ASKED FOR

2. Requirements analysis

The requirements document was broken down into 13 functional modules. Each was given a priority (**M** = must for launch, **S** = should for launch, **C** = could follow after launch) and a complexity rating.

Module	Key requirements (summarised)	Priority	Complexity
Brand & catalog	Categories: Car Scents (hanging diffusers), Car Accessories (ashtrays, front showpieces), Home Scents (room diffusers), Gift Sets. Extensible categories, products, variants.	M	Low
Design & navigation	Premium luxury look, mobile-first, sticky mobile nav, header (logo, menu, search, account, cart), footer, fast media.	M	Medium
Listing pages	2-col mobile grid, cards with price/sale/discount/stock, **swipeable photo+video on cards** with dots/arrows, Quick View, Quick Add, wishlist, search suggestions, sort, filters (price, type, fragrance, availability), load more.	M	High
Product page	Gallery with video, zoom/full-screen, fragrance notes, variants, quantity, stock & delivery info, reviews with customer media, related products, policies.	M	Medium
Cart & checkout	Slide-out cart, no reloads, guest checkout, minimal fields, clear totals, COD, JazzCash/Easypaisa, **verified payment status**, confirmation page, email/SMS/WhatsApp notifications, WhatsApp help.	M	High
Accounts & support	Optional accounts, order history, saved addresses, wishlist, recently viewed, contact, FAQ, newsletter w/ consent, feedback incl. photo/video/voice, moderation.	S	Medium
Gifts & promotions	Gift sets/bundles, gift wrap & notes, discount codes, seasonal banners, cross-sell.	M	Medium
Admin dashboard	KPIs (sales, orders by status, customers, repeat, low stock, returns, AOV, growth), date filters, live graphs, actual vs estimated labels.	M	High
Order management	Auto-saved orders, 8 statuses, search/filter, notes, courier & tracking, invoices & packing slips, CSV/Excel export, refunds/returns history.	M	Medium
Customer CRM	Auto-built customer DB, dedupe, order history, LTV/AOV, first/last purchase, segments (new/repeat/high-value/inactive), notes, consent, export, access control.	M	Medium
Products & inventory	CRUD/duplicate, media upload, fragrance variants, gift sets, stock with auto-deduction, low-stock alerts, cost prices, adjustment ledger, bulk import/export.	M	Medium
Analytics & finance	Sales by period/product/category/fragrance, best & slow sellers, repeat rate, COD vs prepaid, cancellation/delivery rates, coupon performance, gross/net sales, **estimated profit**, period comparison, exports.	M	High
Marketing	Coupons & usage, banners, subscribers, segments, abandoned-cart tracking & recovery, GA4 + Search Console, restock/promote insights, consent-based messaging.	S	Medium
Reviews & support	Moderation queue, customer media, voice notes, complaints with status & notes, spam protection.	S	Medium
Security	Strong owner auth, 2FA, staff roles, activity logs, SSL, updates, automated & tested backups, secure exports, client owns everything.	M	Medium
SEO & platform	SEO URLs/meta, structured data, sitemap, fast mobile, low-cost tools, cost disclosure, handover guide.	M	Low

Business observations

- The market is **Pakistan** (JazzCash, Easypaisa, COD, coco.pk on the packaging). Expect a majority of COD orders, so **order confirmation, fake-order reduction and courier tracking** matter more than card payments.
- Traffic will come mostly from **Instagram/TikTok on mobile data**. Mobile performance and WhatsApp hand-off directly affect conversion.
- The catalogue is small at launch (around 6 fragrances x a few product types), but variants by fragrance are central: fragrance is a first-class entity (shared across diffusers and gift sets, and used for filters and per-fragrance sales reports).
- The owner wants to **run the business from the admin**: CRM, profit and growth reporting are core requirements, not extras.
- Competitors referenced (fieaura.com, colishco.com) use standard Shopify-style themes. Product-card video swiping and a richer fragrance story are realistic ways for COCO to stand out.

CLARIFY BEFORE BUILD

3. Gaps, assumptions & questions for the client

These items are unspecified or ambiguous in the brief. The assumption shown is what this plan uses until the client confirms otherwise.

Topic	Question	Working assumption
Payments	Does COCO already hold JazzCash / Easypaisa **merchant accounts**? (Gateway approval takes 2-6 weeks and needs NTN/business docs.)	Launch with COD + manual wallet transfer (verified in admin); switch on hosted gateways when merchant IDs arrive.
Delivery fees	Flat rate, per-city rates, or free above a threshold?	Flat rate + free-shipping threshold + optional per-city overrides (all editable).
Courier	Which courier(s)? TCS, Leopards, PostEx, Trax, M&P?	Manual tracking number entry at launch; API booking for one courier in a later phase.
Notifications	Email only, or WhatsApp/SMS too? WhatsApp Business API requires Meta verification and is paid per conversation.	Email at launch + click-to-WhatsApp. WhatsApp Cloud API order messages as a phase-2 option.
Stock deduction	Deduct stock when the order is placed or when confirmed?	Configurable; default is on placement, restored on cancel/return.
Gift wrap	Is gift packaging offered, and is it charged?	Optional toggle with configurable price, plus gift note.
Product data	Who supplies photos, videos, descriptions, fragrance notes, cost prices?	Client supplies; developer loads launch catalogue (up to 30 SKUs).
Voice notes	Is voice-note feedback a must for launch?	Phase-2 (browser recording works on most phones, but moderation load is real).
Staff	How many staff users at launch, and what may they see?	Owner only at launch; Manager/Staff roles are built in (staff cannot see costs or profit).
Legal	Return/exchange window, refund method, privacy policy text?	Client provides; developer supplies templates to adapt.
Domain & email	Is coco.pk registered and in whose name?	Must be registered to the client (PKNIC). Business email via Zoho/Google Workspace.
Budget	Preferred hosting model: lowest cost (VPS) or fully managed?	VPS for cost; managed option priced in section 15.

PLATFORM CHOICE

4. Technology stack evaluation & decision

Four realistic options were scored against the brief (5 = best). Weighted toward owner control, total cost over 3 years, Pakistan payment support and fit with the custom requirements.

Criterion	WooCommerce	Shopify	Medusa v2 + Next.js	Next.js + Payload (chosen)
Fit with custom needs (swipe media, CRM, profit, audit)	2 - many plugins	2 - apps / limits	4	5
JazzCash / Easypaisa + payment verification	3 - 3rd-party plugins	1 - not supported natively	4 - custom provider	5 - custom, full control
No-code admin for owner	4	5	3 - commerce admin, CMS separate	5 - generated from schema, customisable
Mobile performance	2	4	5	5
Running cost (3 yrs)	3 - premium plugins	2 - USD plans + apps	4	5
Security surface & maintenance	2 - plugin updates	5 - managed	4	4
Data ownership / portability	4	2	5	5
Time to first version	5	5	3	4
Overall	**Fast start, weak fit**	**Fast start, poor PK fit**	**Strong; more moving parts**	**Best overall fit**

Why Payload over Medusa

Medusa is an excellent commerce engine, but it runs as a separate backend service plus Redis, and its admin covers commerce only (banners, policies, FAQs and reviews still need a CMS). Payload runs *inside* the Next.js app: one deployment, one database, one admin for catalogue, orders, CRM, content and reports. For a single-brand store with about 30-200 SKUs, the commerce primitives we need (cart, pricing, coupons, stock ledger, order states) are a small, well-tested amount of code, and owning them is what makes the custom analytics and PK payment flows straightforward.

WooCommerce: built-in vs plugin (as requested in the brief)

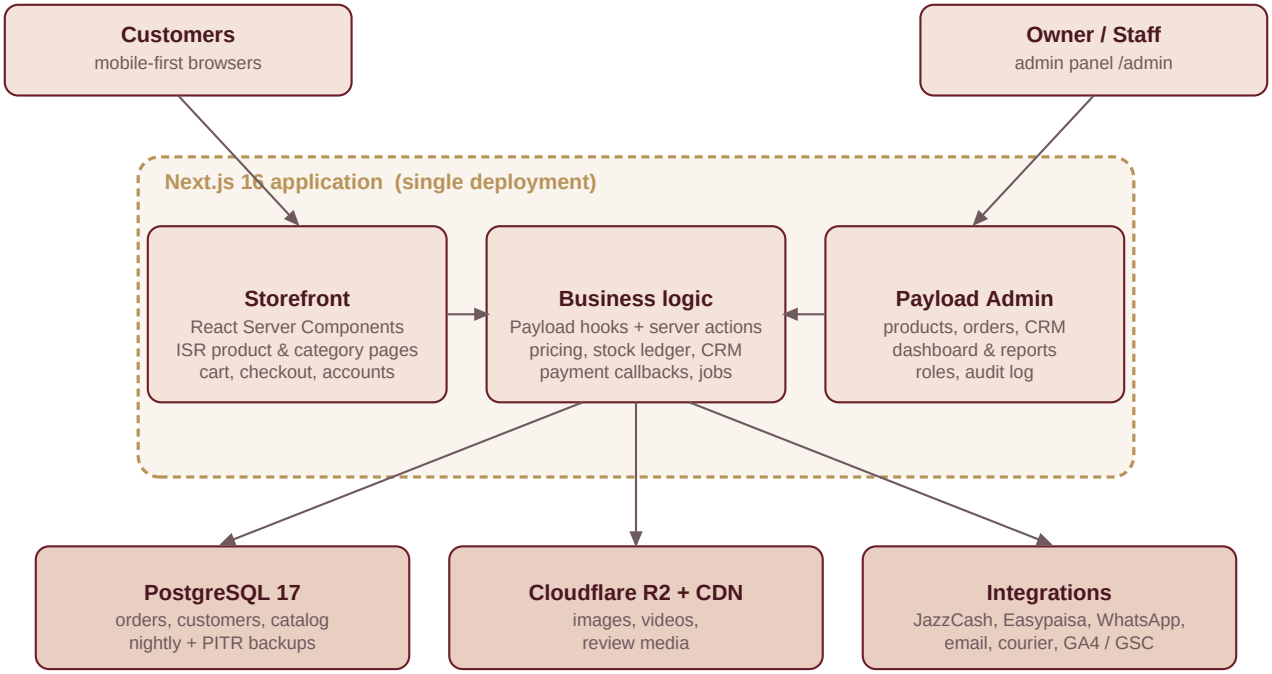
Requirement	WooCommerce status
Products, categories, variations, coupons, COD, guest checkout, basic reports	Built in
Swipeable photo + video on product cards, Quick View, AJAX filters	Premium theme or plugins (e.g. WooCommerce product filters, Quick View) or custom dev
JazzCash / Easypaisa	3rd-party plugins of varying quality; verification logic often weak
Customer CRM with segments, LTV, inactivity	Paid CRM plugin (e.g. Metorik / Jetpack CRM) or custom
Profit reporting with product costs	Paid plugin (Cost of Goods) + custom reports
Review photos/videos, moderation	Paid review plugin
Invoices & packing slips, CSV/Excel export	Free/paid plugins
Abandoned cart recovery	Paid plugin
2FA, activity log, backups	3 separate plugins

Final stack

Layer	Technology	Reason
Frontend	Next.js 16 (App Router, React 19 Server Components), TypeScript	SSR/ISR for SEO and speed; streaming; image optimisation; one language end to end.
Styling / UI	Tailwind CSS 4 + brand tokens; Embla Carousel; Radix primitives	Small CSS, accessible components, smooth touch swiping for cards and galleries.
Admin & backend	Payload CMS 3.9x (inside Next.js)	Auto-generated admin, access control, hooks, uploads, versioning, REST/GraphQL, custom dashboard views.
Database	PostgreSQL 17 (Drizzle via Payload)	Relational integrity for orders/customers; SQL for analytics; mature backups.
Media	Cloudflare R2 + CDN; sharp (WebP); optional Bunny Stream for video	No egress fees, cheap storage, fast delivery of images and short videos.
Email	Resend or Brevo (SMTP) via Payload email adapter	Free tiers cover launch volume; transactional templates.
Payments	COD; JazzCash & Easypaisa hosted checkout with server-side hash verification	Payment marked paid only on verified callback or manual check.
Analytics	GA4 + Search Console; first-party reports from PostgreSQL	Business figures come from our own data, so they always match orders.
Testing / CI	Vitest, Playwright, GitHub Actions	Unit tests for money logic; e2e for checkout and admin flows.
Hosting	Hetzner/DigitalOcean VPS with Coolify (Docker) or Vercel + Neon	Cheap and simple, or fully managed; both keep the same code.

HOW IT FITS TOGETHER

5. System architecture



Single codebase: storefront and admin read and write the same database, so orders and customers appear in the admin instantly.

- **Rendering:** category and product pages are statically generated and revalidated on change (ISR + on-demand revalidation from Payload hooks), so pages load from cache in milliseconds yet reflect price/stock edits immediately.
- **Cart:** client-side store (Zustand, persisted in localStorage) for instant updates; prices and stock are re-validated on the server at checkout.
- **Checkout:** Next.js Server Action -> validates items, recalculates totals, applies coupon, creates the order via Payload Local API inside a database transaction (order + stock ledger + customer upsert commit together).
- **Admin:** Payload admin at /admin with custom React views for the dashboard, reports, invoice printing and exports.
- **Background jobs:** Payload Jobs queue for emails, WhatsApp messages, abandoned-cart reminders, low-stock alerts and nightly report snapshots.
- **Environments:** local (Docker Postgres), staging (client UAT, sandbox payments) and production.

COLLECTIONS ALREADY SCAFFOLDED

6. Data model

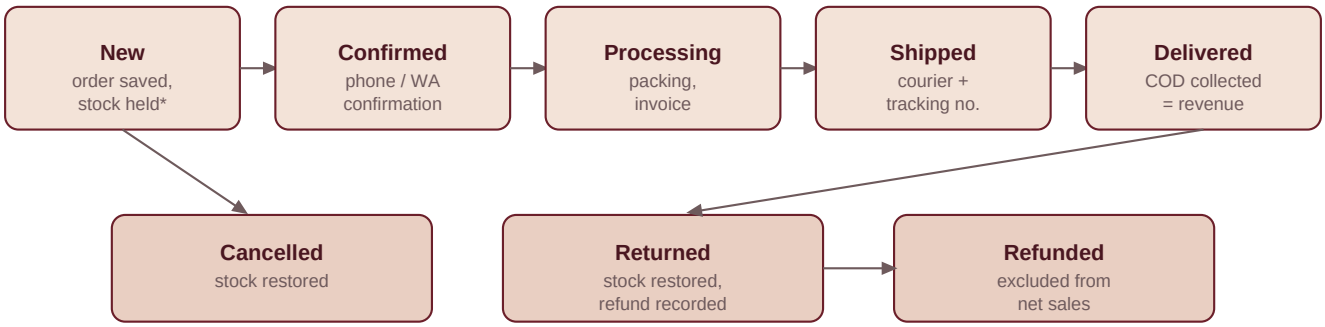
Collection	Purpose & key fields
products	title, slug, type (simple / variable / bundle), gallery (images + videos, ordered), price, salePrice, **costPrice** (hidden from staff), stock, low-stock threshold, fragrance variants (each with sku, price, cost, stock, image), bundle items, specs, features, related products, SEO, status, salesCount, rating.
categories	title, slug, parent (nested), image, sort order, show in menu. Seeded with the client's 4 categories + sub-categories.
fragrances	name, family, mood line, top/heart/base notes, swatch colour. Seeded with Sweet Rose, Citrus Mint, Forest Pine, Vanilla Oud, Lavender Bliss, Cool Ice.
orders	order number, status (8 states) + history, line-item snapshots (title, variant, unit price, unit cost), subtotal/discount/shipping/total, coupon, shipping address, payment (method, status, transaction id, proof, gateway response), fulfilment (courier, tracking, dates), refund, internal notes, gift wrap/note, source.
customers	name, normalised phone (unique dedupe key), email, addresses, marketing consent (email/WhatsApp + timestamp), notes, automatic stats (orders, total spent, AOV, first/last order, segment), joined order list.
inventory-adjustments	Append-only stock ledger: product, variant, change, reason (order, reversal, restock, damage, correction), reference, user.
coupons	code, type (% / fixed / free shipping), value, cap, min order, usage limits, dates, categories, usage count.
banners	hero slides, announcement bar, category banners with desktop/mobile images and schedule.
reviews	product, customer, rating, text, photo/video/voice attachments, verified purchase, status (pending / approved / hidden / spam), admin reply.
support-tickets	type (enquiry, complaint, feedback, return), contact, message, attachments, linked order/customer, status, assignee, resolution notes.
subscribers	email, status, source, consent and unsubscribe timestamps, unsubscribe token.
expenses	non-product costs (ads, packaging, courier...) used in estimated-profit reports.
users / activity-logs	admin users with roles (owner, manager, staff), lockout after failed logins; audit trail of create/update/delete actions.
store-settings (global)	contact & social, shipping rates, payment toggles, gift wrap, stock deduction rule, CRM thresholds, policies, FAQs.
Planned additions	carts (abandoned-cart tracking), customer-accounts (login, linked to customers), wishlists, report-snapshots (daily aggregates).

RULES THE SYSTEM ENFORCES

7. Core business workflows

Order lifecycle

** Owner chooses in Store Settings whether stock is held when the order is placed or when it is confirmed.*



Checkout & payment verification

- Customer submits name, phone, city, address (+ optional email, gift note). Phone is normalised to +923XXXXXXXXX.
- Server recalculates every price from the database, checks stock and coupon rules, then creates the order. Client-side totals are never trusted.
- **COD:** order = New, payment = Pending. Optional COD limit per order. Staff confirm by phone/WhatsApp to reduce fake orders.
- **JazzCash / Easypaisa (gateway):** customer is redirected to the hosted payment page; the gateway's signed callback is verified server-side (hash + amount + order reference) before payment = Paid. Failed or tampered callbacks are logged and ignored.
- **Manual wallet transfer (fallback):** customer may upload a screenshot; payment = *Awaiting verification* until staff confirm the transaction ID in the merchant app. The screenshot alone never marks an order paid (explicit client requirement).
- Confirmation page with order number + summary; email confirmation; click-to-WhatsApp with the order number pre-filled.

Customer database (CRM)

- Each order looks up the customer by normalised phone; if none exists, a profile is created. This removes most duplicates for guest checkout.
- After every order change, the customer's order count, total spent, AOV, first/last order and segment are recalculated (cancelled/returned/refunded orders excluded).
- Segments: New, Repeat (2+ orders), High value (spend above threshold), Inactive (no order for N days). Thresholds are editable in Store Settings.
- Marketing consent is stored with a timestamp; exports and campaigns only include customers who opted in.

Inventory

- Stock is deducted when the order enters a stock-holding status and restored on cancel/return. An idempotency flag prevents double counting.
- Every movement writes to the inventory ledger, so stock levels can always be reconstructed and audited.
- Low-stock alerts on the dashboard and by email when stock falls to the product's threshold.

Financial figures

- **Gross sales** = sum of line items; **Net sales** = gross - discounts - refunds (shipping shown separately).

- **Estimated profit** = net sales - cost of goods (cost snapshot stored on each order line) - recorded expenses. Always labelled 'estimated'.
- Revenue counts delivered and in-progress orders; cancelled, returned and refunded orders are excluded and reported separately.

NOTHING LEFT IMPLICIT

8. Requirement-to-implementation map

Requirement	Implementation	Type
Swipeable photos/videos on product cards	Embla carousel per card, lazy-loaded slides, dot indicators, arrows on desktop; videos are muted, autoplay-on-visible MP4/WebM with poster images.	Custom
Quick View / Quick Add	Modal using the same product component; quick add uses the default variant, or opens a variant picker when a fragrance must be chosen.	Custom
Search with suggestions	PostgreSQL full-text + trigram search over titles, fragrances and categories; debounced suggestions API.	Custom
Filters & sort	URL-driven filters (price, type, fragrance, in stock), sorts (price, newest, popularity = salesCount); server-rendered for SEO.	Custom
Zoom & full-screen gallery	Pinch-zoom lightbox (yet-another-react-lightbox) with video slides.	Library
Wishlist & recently viewed	localStorage for guests, synced to account when signed in.	Custom
Guest checkout + optional accounts	Checkout never requires login. Accounts via email magic link / OTP, linked to the CRM customer.	Custom
Discounts, bundles, gift wrap	Coupons collection + pricing engine (unit-tested); bundle product type; gift wrap price in settings.	Built (scaffolded)
Notifications	Email templates (React Email) via Resend/Brevo; WhatsApp Cloud API templates (optional); SMS via local provider (optional).	Integration
Admin dashboard & reports	Custom Payload admin views: KPI cards, Recharts graphs, date presets + custom range, period comparison, CSV/XLSX export.	Custom
Invoices & packing slips	Printable HTML -> PDF (browser print / server-side) from the order view; bulk print.	Custom
Bulk import/export products	CSV import with validation preview; export to CSV/XLSX.	Custom
Review & feedback moderation	Pending-by-default reviews, media upload limits and type checks, Cloudflare Turnstile anti-spam, rate limiting.	Built (scaffolded) + custom
Voice-note feedback	MediaRecorder API in browser -> audio upload to R2 -> moderation queue.	Phase 2
Abandoned carts	Cart snapshots for visitors who entered a phone/email; reminder via email/WhatsApp after 1h/24h with consent.	Custom (phase 2)
2FA & roles	Owner/manager/staff roles (done); TOTP 2FA for admin login.	Custom
Activity log	Hooks record admin create/update/delete with changed fields (done for products and orders).	Built (scaffolded)
Backups	Nightly pg_dump + WAL/PITR (managed DB) to R2 with 30-day retention; restore drill before launch.	Ops
SEO	Payload SEO plugin, JSON-LD Product/Offer/Review/Breadcrumb, sitemap.xml, robots.txt, canonical URLs.	Built (scaffolded) + custom

THIRD-PARTY SERVICES

9. Integrations

Service	Use	Launch?	Notes
JazzCash (Hosted Checkout)	Mobile wallet & card payments	If merchant account ready	Requires business merchant account; sandbox first; HMAC-SHA256 secure hash verification.
Easypaisa (Easypay)	Mobile wallet payments	If merchant account ready	Requires merchant onboarding; callback/IPN verification and status inquiry.
Cash on Delivery	Default payment method	Yes	Order confirmation step and COD limit to reduce fake orders.
Resend / Brevo	Transactional email	Yes	Free tier covers launch volume; SPF/DKIM on coco.pk domain.
WhatsApp	Help button + order updates	Button yes; API optional	Click-to-chat is free. Cloud API needs Meta Business verification and per-conversation fees.
Couriers (PostEx / Leopards / TCS)	Tracking & booking	Manual tracking at launch	API booking and status sync for one courier in phase 2.
Google Analytics 4 + Search Console	Traffic, conversions, indexing	Yes	E-commerce events: view_item, add_to_cart, begin_checkout, purchase. Consent banner.
Meta Pixel / TikTok Pixel	Ad attribution	Optional	Server-side Conversions API recommended later.
Cloudflare	DNS, CDN, WAF, R2, Turnstile	Yes	Free plan is sufficient.

PROTECTING THE BUSINESS

10. Security, privacy & backups

- **Authentication:** admin login with strong passwords, lockout after 5 failed attempts (configured), secure HTTP-only cookies, 8-hour sessions, TOTP 2FA for owner and managers.
- **Authorisation:** role-based access is enforced on the server for every collection and field. Staff cannot see cost prices, profit, expenses or audit logs; only the owner manages users and settings.
- **Data protection:** HTTPS everywhere (Cloudflare + Let's Encrypt), secrets only in environment variables, customer PII never sent to analytics, exports restricted and audit-logged.
- **Input & upload safety:** server-side validation, MIME/size limits on uploads, Turnstile on public forms, rate limiting on checkout, reviews and login.
- **Payments:** no card data touches COCO servers (hosted checkout); callbacks verified by signature, amount and order reference; idempotent processing.
- **Backups:** nightly database dumps + media bucket versioning, copied off-site with 30-day retention. Restore is tested before launch and quarterly, following a written runbook.
- **Updates:** monthly dependency updates (Dependabot), security patches within 72 hours. Far smaller attack surface than a plugin-based site.
- **Ownership:** domain, hosting, database, Cloudflare, email, payment and Google accounts are registered in the client's name; the developer is added as a user and removed at handover if requested.

FAST ON PAKISTANI MOBILE NETWORKS

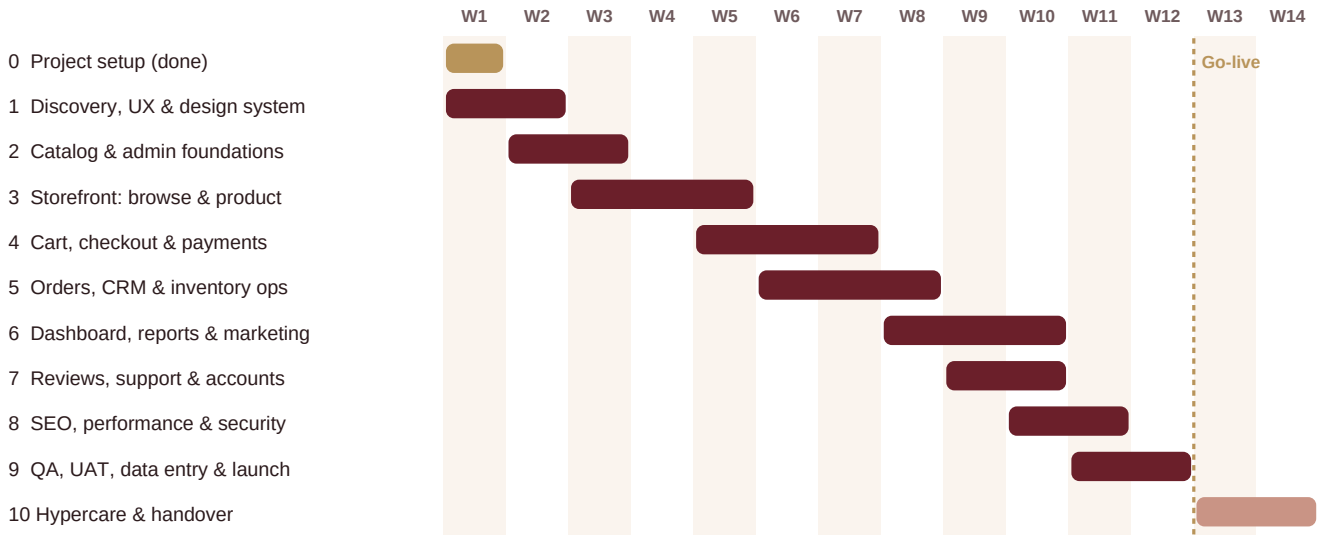
11. SEO & performance

Metric (mobile, 4G throttled)	Target
Largest Contentful Paint	< 2.5 s (aim < 1.8 s)
Interaction to Next Paint	< 200 ms
Cumulative Layout Shift	< 0.1
JavaScript on category page	< 170 KB gzipped
Lighthouse Performance / SEO / Accessibility	90+ / 100 / 95+

- Images: WebP/AVIF in responsive sizes (320 / 640 / 1400 px), lazy-loaded, focal-point cropping, blur placeholders.
- Videos: short (6-15 s), compressed H.264/WebM at 720p and under about 1.5 MB, poster frame, loaded only when the card is visible.
- Server Components + ISR to keep client JavaScript small; fonts self-hosted through next/font.
- SEO: clean URLs (/collections/car-scents, /products/sweet-rose-hanging-diffuser), editable meta titles and descriptions, Open Graph images, JSON-LD (Product, Offer, AggregateRating, BreadcrumbList, Organization), XML sitemap, robots.txt, canonical tags, 301 redirect manager.
- Accessibility: WCAG 2.2 AA contrast with the brand palette, focus states, labelled controls, at least 44 px touch targets, reduced-motion support.

ABOUT 12 WEEKS TO LAUNCH

12. Delivery plan & timeline



Team: 1 senior full-stack developer (lead), UI/UX designer (~25% for weeks 1-4), QA tester (~30% for weeks 9-12). Phases overlap where dependencies allow. A demo is held on staging at the end of every second week.

WHAT GETS BUILT, IN ORDER

13. Sprint-level task breakdown

Phase 0 - Project setup (complete)

- Next.js 16 + Payload 3 + PostgreSQL project, TypeScript strict, ESLint, Prettier, Vitest, Playwright.
- Full data model (15 collections + Store Settings), roles & access control, order hooks (order numbers, CRM upsert, stock ledger, customer stats), activity log.
- Brand tokens (burgundy / blush / cream / gold), fonts, header/footer shell, admin KPI widget, seed script, Docker Postgres, env template.

Phase 1 - Discovery, UX & design system (W1-2)

- Workshop to close the open questions in section 3; confirm payment accounts, courier and delivery fees.
- Wireframes (mobile first): home, category, product card states, Quick View, PDP, cart drawer, checkout, confirmation, account.
- High-fidelity UI in Figma from the brand board; component library (buttons, chips, price, badges, rating, fragrance swatch).

Phase 2 - Catalog & admin foundations (W2-3)

- Admin UX: grouped navigation, list columns, filters, product duplicate action, media guidelines, live preview.
- R2 media storage + image sizes; video upload rules and poster frames; staging environment and CI pipeline.
- Load sample catalogue (6 fragrances x diffusers, accessories, 2 gift sets) for design and testing.

Phase 3 - Storefront browse & product (W3-5)

- Home (hero banners, categories, best sellers, new arrivals, gift sets, reviews strip, Instagram band).
- Category pages: 2-column mobile grid, **swipeable media cards**, filters, sort, load more, empty states.
- Product page: gallery with video + zoom, fragrance selector & notes, stock & delivery info, specs, policies, related products.
- Search with suggestions, Quick View, wishlist, recently viewed, sticky mobile nav, WhatsApp button.

Phase 4 - Cart, checkout & payments (W5-7)

- Slide-out cart, quantity edits, coupon field, free-shipping progress bar, gift wrap/note.
- One-page guest checkout; server-side pricing; order creation in a transaction; confirmation page.
- COD flow and limits; JazzCash and Easypaisa hosted checkout + callback verification (sandbox to live); manual transfer fallback.
- Order emails (confirmation, shipped, delivered); WhatsApp click-to-chat with order number.

Phase 5 - Orders, CRM & inventory operations (W6-8)

- Order list views by status, quick status actions, courier/tracking entry, internal notes, refunds/returns.
- Invoice and packing slip printing (single and bulk); CSV/XLSX export for orders, customers and products.
- Customer profile view (history, LTV, segment, consent), segment filters, notes; dedupe merge tool.
- Manual stock adjustments, low-stock alerts, bulk product CSV import/export with preview.

Phase 6 - Dashboard, reports & marketing (W8-10)

- Dashboard: KPI cards, order-status funnel, revenue and orders graphs, date presets, custom range and comparison.
- Reports: sales by product/category/fragrance, best and slow sellers, AOV, repeat rate, COD vs prepaid, cancellation/return/delivery rates, coupon performance, inventory.
- Finance: gross, discounts, refunds, net, COGS, expenses, **estimated profit** with clear labels; exports.
- Marketing: coupons UI, banners scheduling, subscribers and unsubscribe, abandoned-cart capture.

Phase 7 - Reviews, support & accounts (W9-10)

- Review submission with photo/video, moderation queue, rating aggregation on products, verified-purchase badge.
- Contact form and complaints to support tickets with status and resolution notes; FAQ and policy pages from settings.
- Optional customer accounts (OTP / magic link), order history, saved addresses, wishlist sync.

Phase 8 - SEO, performance & security hardening (W10-11)

- Structured data, sitemap, redirects, OG images; GA4 e-commerce events; Search Console verification.
- Performance budget enforcement, image/video audits, Lighthouse CI.
- 2FA, rate limits, Turnstile, security headers (CSP, HSTS), dependency audit; backup automation + restore drill.

Phase 9 - QA, UAT & launch (W11-12)

- Cross-device testing (real Android/iOS phones, Chrome, Safari, Edge, Firefox), accessibility pass, regression e2e suite.
- Client UAT on staging with acceptance checklist; final catalogue entry; live payment test transactions.
- DNS cut-over, monitoring (uptime, error tracking with Sentry), go-live checklist, admin training session.

Phase 10 - Hypercare & handover (W13-14)

- Bug-fix priority window, analytics review after 2 weeks of real traffic, handover pack (section 17).

DEFINITION OF DONE

14. Testing & acceptance

- **Unit tests** (Vitest) for all money and rule logic: pricing, coupons, shipping, stock holding, phone normalisation, segments. 18 tests already pass.
- **Integration tests** against a real PostgreSQL: order creation transaction, stock ledger, CRM stats, payment callback verification.
- **End-to-end tests** (Playwright, mobile and desktop viewports): browse, swipe card media, Quick View, add to cart, checkout (COD and gateway sandbox), admin order update, export.
- **Report reconciliation:** dashboard figures are compared against raw SQL for fixed date ranges (client requirement: graphs must match saved data).
- **Manual QA:** real devices, slow-network throttling, accessibility (axe + keyboard), content proofing.

Client acceptance item	Verified by
Browse all categories on mobile and desktop	E2E + manual device matrix
Swipe photos/videos on product cards	E2E (touch emulation) + real phones
Quick View and Quick Add work	E2E
Checkout works and orders are saved accurately	E2E + integration test of totals
Owner can find and update orders	E2E admin flow + UAT
Customer profiles & histories saved and searchable	Integration + UAT
Products, media, prices, stock editable in admin	UAT
Dashboard graphs match saved data	Reconciliation script vs SQL
Sales and profit calculated & labelled correctly	Unit tests + reconciliation
Reports can be exported	E2E download test
Backups and recovery tested	Restore drill, signed off in runbook
Paid tools, limits and costs disclosed	Section 15 + handover pack
Tested on real mobiles and common browsers	QA matrix report

FULL DISCLOSURE, AS REQUESTED

15. Costs & recurring fees

Indicative prices as of October 2026; confirm at signing. No paid plugins or licences are required: Next.js, Payload, PostgreSQL and the libraries used are open source (MIT/PostgreSQL licences).

Item	Option A: lowest cost	Option B: fully managed	Notes
Application hosting	VPS 2-4 vCPU / 4-8 GB (Hetzner, DigitalOcean): USD 6-25 / mo	Vercel Pro: USD 20 / mo per seat	Option A runs app + DB via Coolify/Docker.
Database	Included on VPS (+ off-site backups)	Neon / Supabase: USD 0-25 / mo	Managed DB adds point-in-time recovery.
Media storage (R2)	Free up to 10 GB, then USD 0.015 / GB	Same	No bandwidth (egress) charges.
Domain coco.pk	PKNIC: approx. PKR 2,500-3,000 / 2 yrs	Same	Must be in client's name.
Cloudflare (DNS, CDN, SSL, WAF)	Free	Free	
Transactional email	Resend/Brevo free tier (about 3,000 / mo)	Same	Paid plans from about USD 20 / mo if volume grows.
Business mailbox	Zoho Mail free / Google Workspace about USD 7 / user / mo	Same	Optional.
WhatsApp Cloud API	Click-to-chat: free	Template messages: per-conversation Meta fees	Only if automated WhatsApp messages are wanted.
Error monitoring / uptime	Sentry free + UptimeRobot free	Same	
Payment gateways	JazzCash / Easypaisa: per-transaction MDR set in merchant agreement	Same	Rates are negotiated; typically a low single-digit percentage. Get written quotes.
Courier	Per-parcel and COD handling fees per courier contract	Same	Outside the website budget.
Typical monthly total	**about USD 10-30**	**about USD 45-70**	Excludes gateway and courier fees.

Development & maintenance
Development cost is quoted separately per phase. Maintenance retainer (optional): updates, backups monitoring, minor changes. Typically 6-10 hours per month.

WHAT COULD GO WRONG

16. Risks & mitigations

Risk	Impact	Mitigation
Merchant accounts for JazzCash/Easypaisa delayed	High	Start applications in week 1; launch with COD + verified manual transfer; gateway code built against sandbox.
Product photos/videos not ready	High	Shot list and specs delivered in week 1; placeholder catalogue for development; content freeze in week 10.
Fake / unconfirmed COD orders	Medium	Confirmation workflow, COD limit, phone validation, blacklist flag on customer profile.

Risk	Impact	Mitigation
Scope creep from 'nice-to-haves'	Medium	MoSCoW priorities (section 2), change requests estimated separately, phase-2 backlog.
Heavy videos slow mobile pages	Medium	Size limits and compression guide, lazy loading, poster-first, performance budget in CI.
Single-developer dependency	Medium	Typed codebase, README and architecture docs, CI tests, client owns repo and all accounts.
Data loss	High	Nightly off-site backups, tested restores, append-only stock ledger and audit logs.
Framework upgrades (Payload 4 currently in beta)	Low	Launch on stable Payload 3.x; plan upgrade after 4.0 is GA and stable; pinned versions.

CLIENT INDEPENDENCE

17. Handover, ownership & maintenance

- **Owner manual** (PDF + short screen recordings): add/edit products and fragrances, upload media, manage stock, process orders, print invoices, issue refunds, manage coupons and banners, moderate reviews, read reports, export data.
- **Operations runbook:** deployment, environment variables, backup and restore procedure (tested), rotating credentials, adding staff users, incident checklist.
- **Asset register:** every account (domain, hosting, Cloudflare, R2, email, Google, Meta, payment gateways, GitHub) with owner, login holder and renewal date.
- **Source code** in the client's GitHub organisation; developer access removable at any time.
- **Disclosure list** of all third-party services, limitations and recurring costs (section 15, kept up to date).
- One live training session (60-90 min) for the owner and staff, plus 2 weeks of hypercare after launch.

ALREADY IN THIS REPOSITORY

18. Project setup completed

Area	What is in place
Framework	Next.js 16.3 + React 19.2 + Payload 3.92 + PostgreSQL adapter, TypeScript strict; type check passes.
Data model	15 collections + Store Settings global (section 6), generated TypeScript types, admin grouped into Sales, Catalog, Marketing, Customers & Support, Reports, Settings.
Business logic	Pricing engine, PK phone normalisation, order-status rules, customer segmentation (src/lib); order hooks for numbering, CRM upsert, stock ledger, customer stats; activity logging.
Access control	Owner / manager / staff roles; cost and profit fields hidden from staff; public reads limited to published products and approved reviews.
Admin	KPI widget above the dashboard (30-day revenue, orders, AOV, new orders, customers, live products).
Storefront shell	Tailwind 4 with COCO brand tokens, Cormorant Garamond + Jost fonts, header, footer and home page reading categories.
Plugins	SEO plugin (products and categories), S3/R2 storage (enabled by environment variable).
Tooling	ESLint, Prettier, Vitest (18 unit tests passing), Playwright, Docker Compose for Postgres, seed script (categories + 6 fragrances), .env.example.
Docs	README, this plan (regenerate with npm run plan:pdf), client requirements and brand references in docs/.

Next steps
Install Node.js 24.22+ (Payload 3.92 requires it), provide a PostgreSQL database (Docker or Neon), then run `npm run dev` and `npm run seed`. Close the questions in section 3 with the client and begin Phase 1.